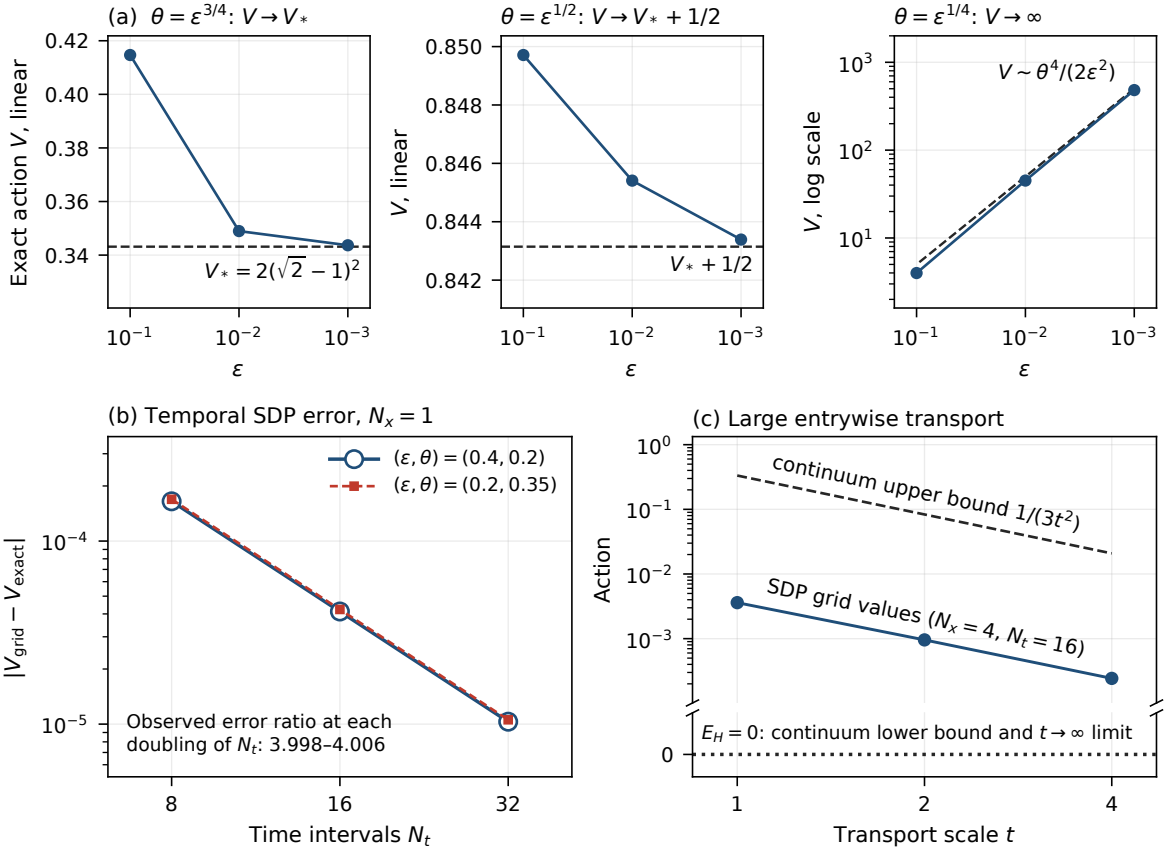

(a) $\theta = \varepsilon^{3/4}$: $V \to V_*$
Exact action V, linear
0.42
0.40
0.38
0.36
0.34
$V_* = 2(\sqrt{2}-1)^2$
10^{-1}
10^{-2}
10^{-3}
ε
$\theta = \varepsilon^{1/2}$: $V \to V_* + 1/2$
V, linear
0.850
0.848
0.846
0.844
0.842
$V_* + 1/2$
$\theta = \varepsilon^{1/4}$: $V \to \infty$
V, log scale
10^3
10^2
10^1
$V \sim \theta^4/(2\varepsilon^2)$
(b) Temporal SDP error, $N_x = 1$
$(\varepsilon, \theta) = (0.4, 0.2)$
$(\varepsilon, \theta) = (0.2, 0.35)$
$|V_{\text{grid}} - V_{\text{exact}}|$
10^{-4}
10^{-5}
Observed error ratio at each
doubling of N_t: 3.998–4.006
8
16
32
Time intervals N_t
(c) Large entrywise transport
Action
10^0
10^{-1}
10^{-2}
10^{-3}
continuum upper bound $1/(3t^2)$
SDP grid values ($N_x = 4$, $N_t = 16$)
$E_H = 0$: continuum lower bound and $t \to \infty$ limit
0
1
2
4
Transport scale t